

PORTFOLIO PROJECT · TEXT ANALYSIS WITH LLM

TaskFlow Support Intelligence System

AI-powered triage pipeline using Groq API & LLaMA 3.3 70B
to classify, prioritise, analyse and respond to support tickets at scale.

PREPARED BY

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BACKGROUND

Product Specialist & QA (SaaS) → Data Analytics

DATASET

50 Tickets · Single Day

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Problem & Approach

THE PROBLEM

TaskFlow faces growing support ticket volume.

- Manual triaging is slow and inconsistent
- No standardised priority assignment
- First responses take too long to draft
- No systematic sentiment tracking
- Escalation risks go undetected

THE SOLUTION

5-prompt LLM pipeline via Groq API

Category Classification	Zero-shot · Temp 0
Priority + ETA	Zero-shot · Temp 0
Sentiment Analysis	Zero-shot · Temp 0
Tags + Summary	Zero-shot · Temp 0
First Response	One-shot · Temp 0.7

All prompts follow the BRIDGE framework · Temperature=0 for consistency · Temperature=0.7 for human-sounding responses

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Key Metrics

One day of support tickets — automatically processed end-to-end

50

TICKETS PROCESSED

In a single day

22%

ESCALATION RATE

11 high-risk tickets flagged

74%

NEGATIVE SENTIMENT

Frustrated + Negative
combined

#1

TOP CATEGORY

Technical Support & Bugs

PRIORITY BREAKDOWN

P1 Critical



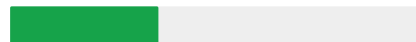
13 tickets

P2 High



16 tickets

P3 Normal



7 tickets

P4 Low



14 tickets

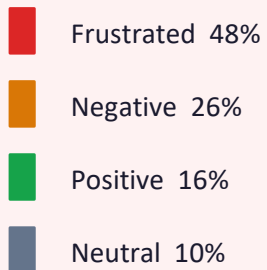
P1 + P2 = 29 tickets (58%) — majority of issues are high severity

Analyst Insights

SENTIMENT

74%

of customers expressed negative or frustrated sentiment on this day.



Core Product Instability

Technical Support & Bugs dominates (22 tickets, 44%). P1+P2 account for 58% — core features are degraded. Engineering intervention is urgent.

Revenue & Churn Risk

Billing & Payment tags alongside Frustrated sentiment = direct churn risk. Same-day resolution required to prevent payment disputes.

Unmet Product Needs

Feature Requests is the 2nd largest category. High volume signals a gap between customer expectations and product capabilities.

Documentation Gap

How-to & Guidance tickets are fully deflectable via a knowledge base. Proper KB documentation would eliminate this category entirely.

■ 22% Escalation Rate — 11 tickets flagged as high churn-risk (Frustrated/Negative + P1 priority)

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Recommendations

01

ENGINEERING · URGENT

Root Cause Analysis on Core Features

Identify and fix non-negotiable buggy features before next release. 22% escalation rate is a direct consequence.

02

PRODUCT · STRATEGIC

Competitor Analysis for Feature Gaps

High Feature Request volume signals competitive risk. Prioritise enterprise customers' requests — retention + revenue opportunity.

03

CUSTOMER SUCCESS · IMMEDIATE

Proactive Customer Communication

Assign a dedicated team to update all open ticket holders. Silence amplifies frustration with 74% negative sentiment.

04

L&D · MEDIUM TERM

Knowledge Base Overhaul

How-to & Guidance tickets are fully deflectable. Structured KB with integration docs eliminates this category entirely.

05

FINANCE · EOD

Billing Issue Same-Day Resolution

Billing disputes with negative sentiment must resolve same-day. Fastest path to involuntary churn if unaddressed.

Fix core bugs · Communicate proactively · Build knowledge base · Resolve billing same-day → Move sentiment from 74% negative to positive baseline
github.com/priyanka-manoj/Text-Analysis-LLM-Taskflow